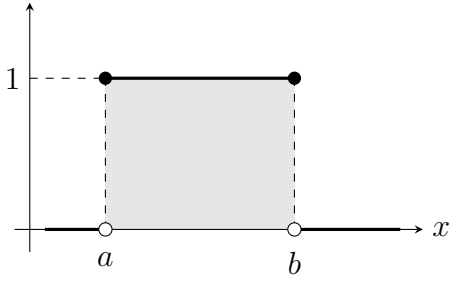
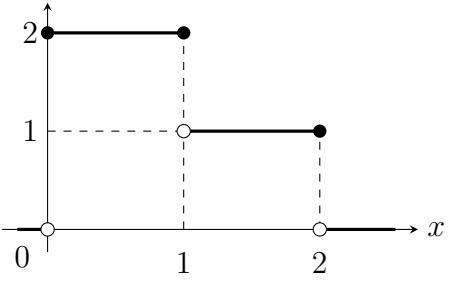
$\mathbf{1}_{[a,b]}$



$2\mathbf{1}_{[0,1]} + \mathbf{1}_{(1,2]}$